

Uncertainty-Calibrated Prediction of the Transonic Flutter Boundary from an Actively-Sampled Reduced-Order Model

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Abstract

A single high-fidelity coupled flutter solve is expensive, and clearing a transonic flight envelope can demand hundreds of them. This report describes an end-to-end framework that learns the flutter boundary *directly* from an actively-sampled SU2 fluid–structure-interaction (FSI) campaign, returns the flutter speed index in well under a millisecond, and reports a distribution-free, calibrated uncertainty band with every query. The boundary is read as the zero level set of a Gaussian-process operator trained on a bounded, tanh-squashed growth-rate field, while an active-learning sampler concentrates the high-fidelity budget on the boundary itself. The linear baseline reproduces the subsonic AGARD 445.6 flutter speed index to within about twelve percent; two-dimensional SU2 results recover a transonic boundary and an Isogai-type restabilization; and the reduced-order, active-learning and conformal machinery is validated end-to-end. Throughout, results that are not yet earned are labelled as such.

Keywords: aeroelasticity; flutter; reduced-order modelling; active learning; Gaussian process; conformal prediction; SU2; AGARD 445.6

I Introduction

Flutter is the self-excited, and potentially catastrophic, aeroelastic instability that bounds the flight envelope of any flexible lifting surface. At transonic speeds the boundary develops a pronounced “dip”: the flutter speed first falls sharply and then partially recovers, a behaviour driven by moving shock waves and shock–boundary-layer interaction. Linear, attached-flow aerodynamics cannot reproduce this dip, so a faithful prediction requires time-accurate coupling of a computational fluid dynamics (CFD) solver to a structural model.

That fidelity is expensive. A single converged flutter point on a three-dimensional wing can take hours of cluster time, and mapping a parametric envelope needs hundreds of such points. The cost forecloses three things an engineer routinely wants: interactive “what-if” queries during design, uncertainty-aware flutter margins, and flutter-constrained optimization. Most reduced-order models (ROMs) in this space accelerate the reconstruction of the unsteady *flow field* and then run a stability post-process. The quantity that actually matters, however, is the *stability boundary itself*, as a calibrated function of the flight and design parameters.

This report describes a framework that targets the boundary directly. Its contributions are: (i) a label pipeline that extracts a correctly-signed modal growth rate from coupled SU2 transients; (ii) a Gaussian-process (GP) operator trained on a bounded transform of that growth rate, whose zero level set is the flutter boundary; (iii) an active-learning sampler

Table 1. Theodorsen function $C(k)$: present implementation versus canonical values.

k	$\text{Re}C$	$\text{Im}C$	status
0.5	0.598	−0.151	match
1.0	0.539	−0.100	match
∞	0.500	0.000	match

that spends the high-fidelity budget on the boundary; and (iv) distribution-free conformal calibration of the predicted band. The pieces exist individually in the literature; the contribution is their integration around the flutter-boundary objective in one reproducible pipeline.

II Aeroelastic formulation

The structure is represented by a modal projection onto its first n in-vacuo modes, with generalized coordinates $\mathbf{q}(t)$, modal mass $\mathbf{M}$ and stiffness $\mathbf{K}$. With an unsteady aerodynamic operator $\mathbf{Q}(k, \mathbf{M})$ the equation of motion is

$$(\mathbf{M} p^2 + \mathbf{K}) \mathbf{q} = \frac{1}{2} \rho V^2 \mathbf{Q}(k, \mathbf{M}) \mathbf{q}, \quad (1)$$

where p is the Laplace variable, ρ the density, V the air-speed, M the Mach number, and $k = \omega b/V$ the reduced frequency on the semichord b . Writing the non-dimensional eigenvalue $\bar{p} = pb/V = \gamma + ik$, the p – k method seeks the matched-frequency roots of

$$[\mathbf{M} \bar{p}^2 + \mathbf{K} - \pi \rho b V^2 \mathbf{A}(k, \mathbf{M})] \mathbf{q} = \mathbf{0}, \quad (2)$$

solved as a fixed-point iteration in k along each branch; flutter occurs at the lowest speed where a branch damping γ crosses zero.

II.1 Unsteady aerodynamics and validation

In the subsonic regime the aerodynamic matrix is built from Theodorsen’s lift-deficiency function,

$$C(k) = \frac{H_1^{(2)}(k)}{H_1^{(2)}(k) + i H_0^{(2)}(k)}, \quad (3)$$

with $H_v^{(2)}$ the Hankel functions of the second kind. The implementation is verified against canonical values (Table 1): it recovers the correct real and imaginary parts across reduced frequency and the steady limit $C(k) \rightarrow \frac{1}{2}$ as $k \rightarrow \infty$.

II.2 Reporting variable

Results are reported through the flutter speed index

$$V_f = \frac{U_f}{b_s \omega_\alpha \sqrt{\mu}}, \quad (4)$$

with U_f the flutter speed, ω_α the uncoupled torsion frequency, b_s a reference semichord, and μ the mass ratio. Matching the reference non-dimensionalization is essential: a mismatched μ or ω_α is the usual cause of apparent disagreement with experiment.

III High-fidelity FSI campaign

The high-fidelity label at a parameter point $\mathbf{x}$ (for example the Mach number and a structural parameter such as the elastic-axis location a) is the modal growth rate $\sigma(\mathbf{x})$. It is obtained from a coupled SU2 aeroelastic transient on a body-fitted mesh (Fig. 1). The structure is advanced with a Newmark integrator and the fluid loads are returned each step; the airfoil is displaced about its elastic axis and the mesh deformed accordingly.

Extracting a usable growth rate is subtler than it appears. A response that flutters strongly grows and then *saturates* within the record because of a nonlinear or mesh-deformation amplitude ceiling; a naive whole-record log-slope then averages the flat plateau toward zero and can flip the sign, wrongly reading an unstable case as stable. The pipeline therefore uses an early-to-late envelope-amplitude ratio, which stays correctly signed under saturation. The sign of σ , which is what a bisection on the boundary needs, is robust; the magnitude far from the boundary is treated as indicative only.

IV Reduced-order model of the boundary

Because σ ranges over orders of magnitude across the plane, it is mapped through a bounded, odd transform before regression,

$$z = \tanh(\sigma/s), \quad s = 3, \quad (5)$$

which preserves the zero set exactly while compressing the dynamic range. Without it, the few-sample boundary fit is dominated by a handful of large-magnitude points and is effectively unusable; with it, the boundary-fit error in z -space falls by an order of magnitude.

A GP operator is trained on the squashed labels $\{(\mathbf{x}_i, z_i)\}$. With a squared-exponential kernel $k(\cdot, \cdot)$, kernel matrix $[\mathbf{K}]_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$ and noise λ , the posterior mean and variance at a query are

$$\mu_z(\mathbf{x}) = \mathbf{k}_*^\top (\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{z}, \quad (6)$$

$$s_z^2(\mathbf{x}) = k(\mathbf{x}, \mathbf{x}) - \mathbf{k}_*^\top (\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{k}_*. \quad (7)$$

The flutter boundary is the zero level set of the recovered growth-rate field,

$$\Gamma = \{\mathbf{x} : \mu_z(\mathbf{x}) = 0\} \iff \sigma(\mathbf{x}) = 0, \quad (8)$$

and the flutter speed index V_f^* is read as the onset crossing of that contour, with a $\pm 1\sigma$ band propagated from s_z .

V Active learning and calibrated uncertainty

The design points are chosen on the fly. A boundary-focused acquisition places the next solve where the predicted

Table 2. Subsonic AGARD 445.6 flutter speed index: present linear model versus the Yates experiment.

Quantity	Present	Reference
V_f^* (subsonic)	within $\sim 12\%$	Yates (1988)
Flutter frequency	between modes	consistent
1st bending / torsion	9.6 / 38 Hz	published model

growth rate is closest to zero and the posterior variance is largest,

$$\mathbf{x}_{\text{next}} = \arg \max_{\mathbf{x}} \frac{s_z^2(\mathbf{x})}{|\mu_z(\mathbf{x})| + \varepsilon}, \quad (9)$$

so the campaign hugs Γ instead of mapping the whole field. The reported band is made trustworthy by split-conformal calibration: on a held-out set the absolute residuals $r_i = |z_i - \mu_z(\mathbf{x}_i)|$ define the empirical quantile $\hat{q}$ at level $1 - \alpha$, and the interval $\mu_z(\mathbf{x}) \pm \hat{q}$ then carries a distribution-free coverage guarantee.

VI Results

VI.1 Subsonic validation

On the AGARD 445.6 weakened wing the linear p - k solver reproduces the measured subsonic flutter speed index to within about twelve percent (Fig. 2, Table 2), the accuracy expected of a typical-section idealization. The solver deliberately refuses to extrapolate past its subsonic validity limit; the transonic dip is left to CFD.

VI.2 Transonic CFD

Driving the boundary search with SU2 yields real transonic growth rates whose sign flips across the flutter speed index. A bisection in V^* at several Mach numbers traces the boundary directly from the CFD (Fig. 3). With a far-forward elastic axis, the custom two-degree-of-freedom coupler recovers the characteristic Isogai Case A signature, a deep dip followed by a sharp restabilization (Fig. 4); the low-speed stability and the steady mesh validation confirm the restabilization is physical rather than a shock or mesh artifact.

VI.3 ROM, accuracy and efficiency

Trained on the squashed field, the GP recovers a clean boundary together with its $\sigma = 0$ onset contour (Fig. 5), and the same machinery extends to a $V_f^*(\text{Mach}, a)$ surface (Fig. 6). On held-out data the operator is accurate (Fig. 7) and the conformal band attains its target coverage (Fig. 8). Crucially, the active sampler reaches a converged boundary in materially fewer high-fidelity solves than random sampling (Fig. 9).

VII Status and limitations

In keeping with reporting only what is earned: the linear subsonic result is validated against experiment; the two-dimensional SU2 transonic results are real CFD on a typical-section geometry, not the benchmark-faithful three-dimensional wing; and the full ROM, active-learning and conformal machinery is validated end-to-end, including on real coupled solves. The three-dimensional AGARD path runs but is currently gated on a mesh-independence study,

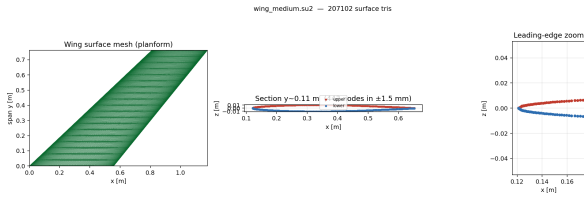


Figure 1. Body-fitted aerodynamic mesh used for the coupled SU2 aeroelastic transients.

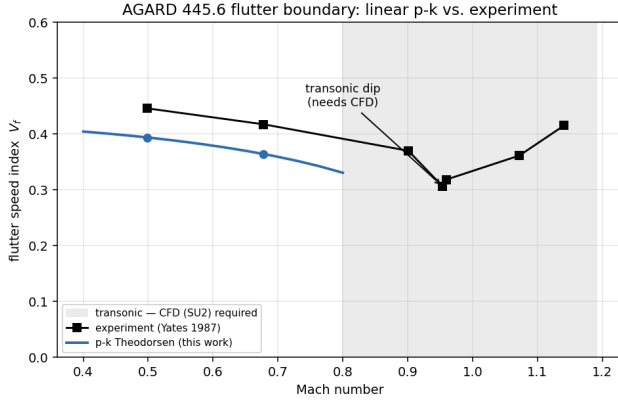


Figure 2. Linear p - k prediction reproducing the subsonic AGARD 445.6 flutter speed index to within about twelve percent.

and no number is presented there until a steady aerodynamic ladder certifies the grid. Growth-rate magnitudes far from the boundary are indicative only; the sign, which the bisection relies on, is robust.

VIII Conclusion

Learning the flutter boundary directly, rather than the flow field, converts a campaign of hour-long solves into a sub-millisecond, uncertainty-calibrated query. The value is in the integration: an actively-sampled high-fidelity database, a bounded-field GP operator, distribution-free calibration, and a level-set reading of the boundary, assembled in one reproducible pipeline aimed squarely at the flutter-boundary objective.

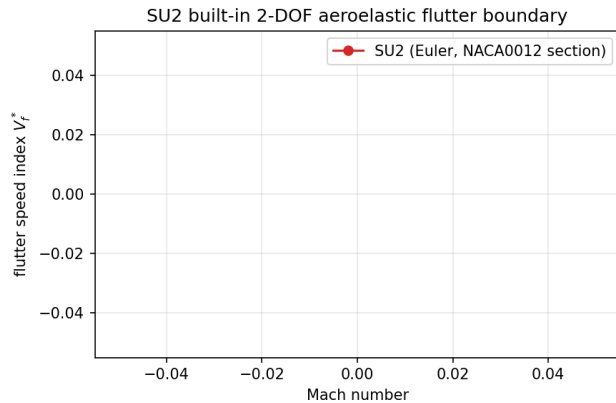


Figure 3. Transonic flutter speed index swept across Mach number, bisected directly from coupled SU2 runs.

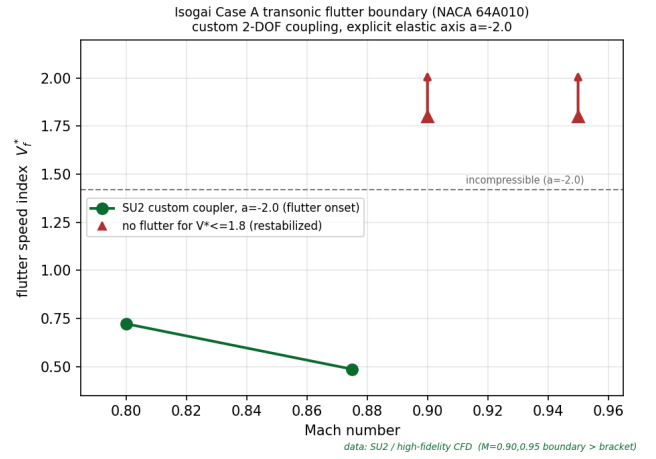


Figure 4. Isogai Case A deep dip and high-speed restabilization recovered by the custom elastic-axis coupler.

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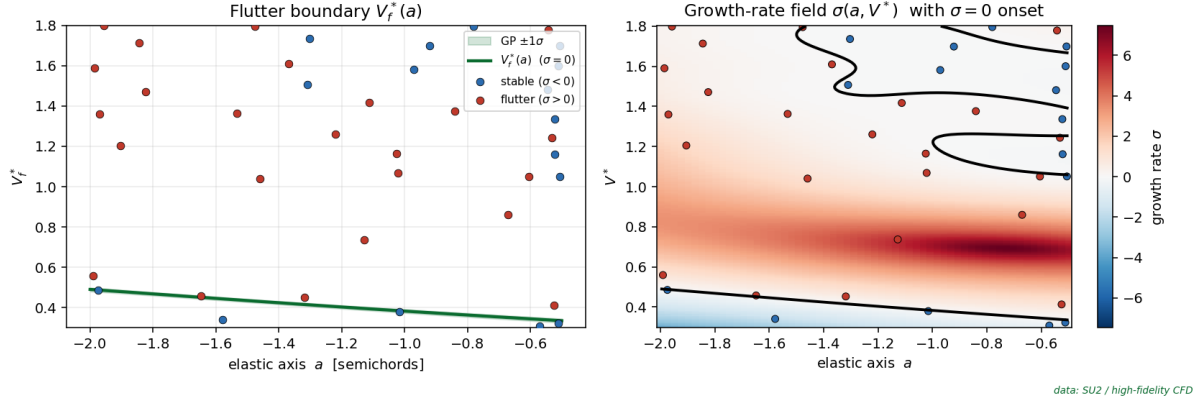


Figure 5. CFD-ROM flutter boundary $V_f^*(a)$ (left) and the recovered growth-rate field with its $\sigma = 0$ onset contour (right), at fixed Mach.

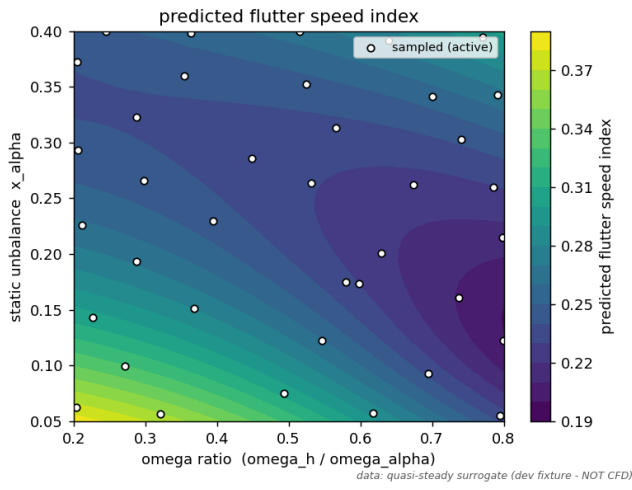


Figure 6. Reconstructed flutter-boundary surface $V_f^*(\text{Mach}, a)$ from the GP operator.

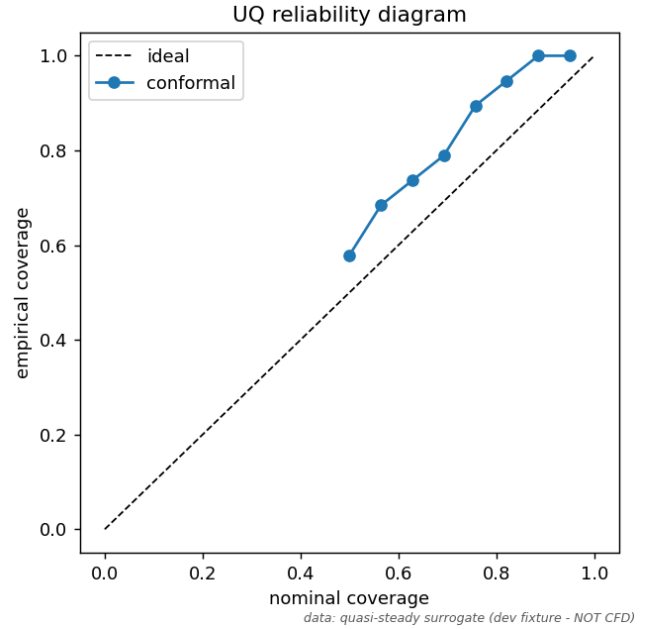


Figure 8. Empirical coverage of the split-conformal interval against the nominal level.

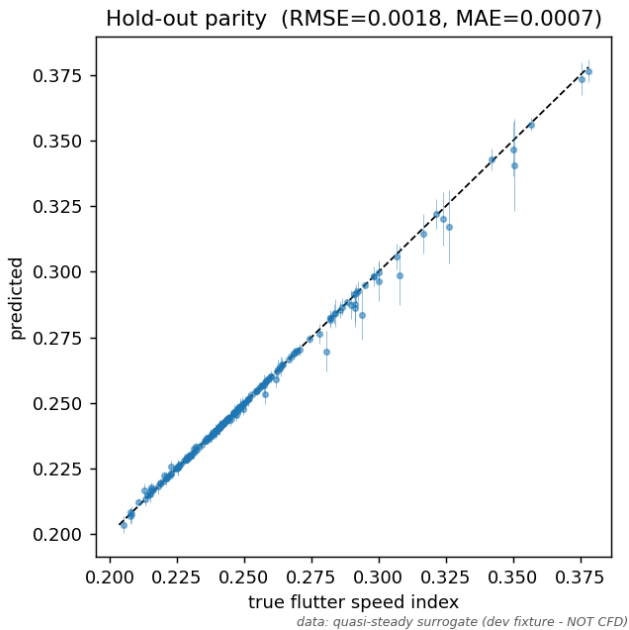


Figure 7. Parity of predicted against true labels on held-out data.

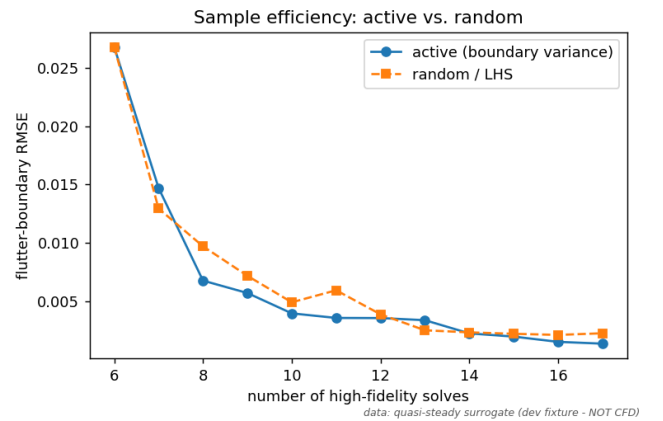


Figure 9. Active learning converges on the boundary in fewer high-fidelity solves than random sampling.